EXPERIMENT 7

Aim: Configuring a Cryptocurrency Wallet using Metamask and integrating it with a DApp

THEORY

1. Cryptocurrency Wallet

A cryptocurrency wallet is a digital application used to manage blockchain accounts. It stores private keys that allow users to access their cryptocurrency and authorize transactions.

2. MetaMask

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with Ethereum networks and decentralized applications (DApps). It manages accounts, displays balances and signs transactions.

3. Ethereum Sepolia Test Network

Sepolia is an Ethereum test network used for testing blockchain applications and transactions without spending real ETH. Test ETH can be obtained through faucets.

4. Ethereum Wallet Address

A wallet address is a unique identifier of an Ethereum account. It is used to send and receive ETH and interact with smart contracts.

5. Decentralized Application (DApp)

A DApp is an application that interacts with blockchain technology. It uses a frontend interface, a Web3 provider and a blockchain network to perform decentralized operations.

6. Wallet–DApp Integration

Wallet–DApp integration allows users to connect their MetaMask wallet to a web application. After permission is granted, the DApp can access the wallet address and request blockchain transactions.

7. Ethers.js

Ethers.js is a JavaScript library used to connect web applications with Ethereum. It provides functions to access wallet accounts, retrieve balances and send transactions.

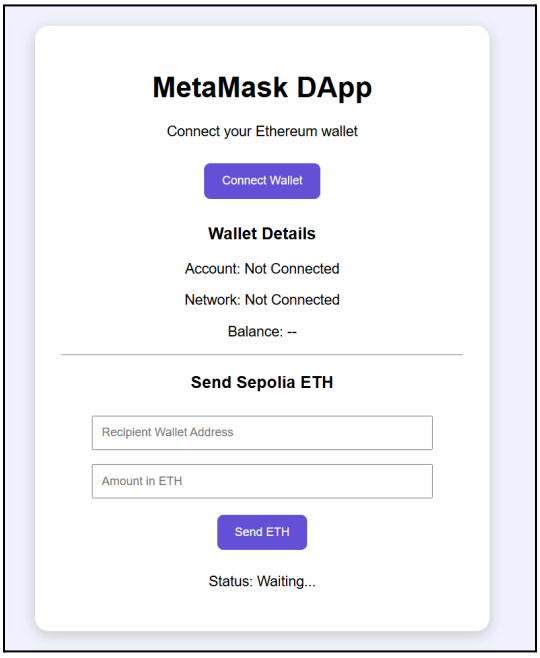
8. Gas Fees and Transactions

Gas fees are the transaction costs required to perform operations on the Ethereum network. MetaMask allows users to review and approve transactions before they are submitted to the blockchain.

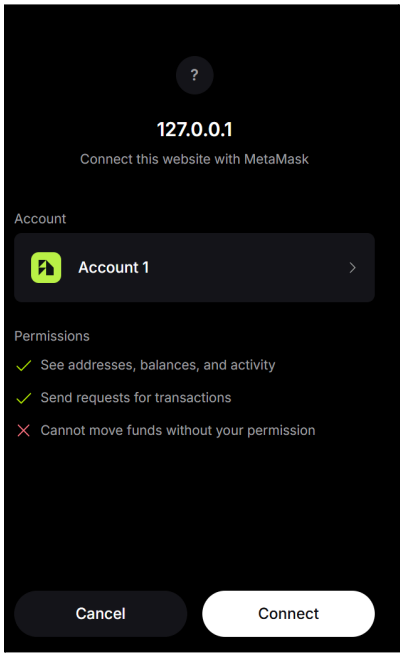
9. Blockchain Transaction Verification

Transactions submitted to Ethereum are recorded on the blockchain. Sepolia Etherscan can be used to verify transaction details such as sender, receiver, amount and transaction status.

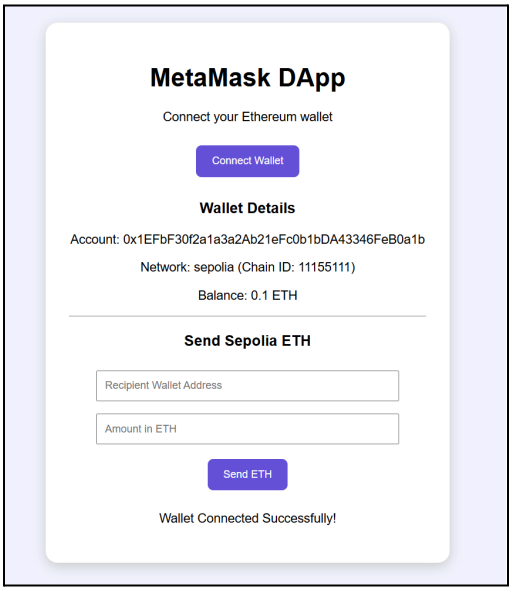
Output & Code:



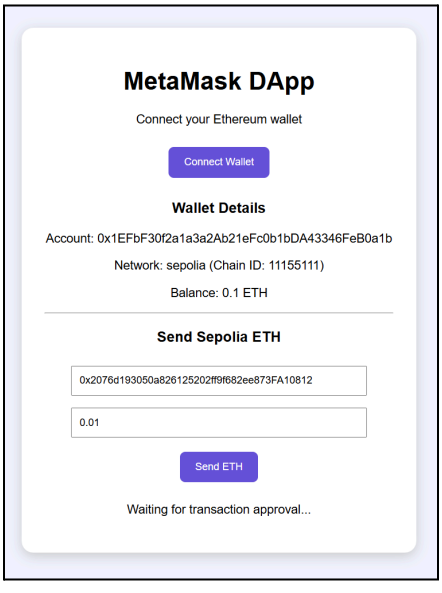
MetaMask DApp interface



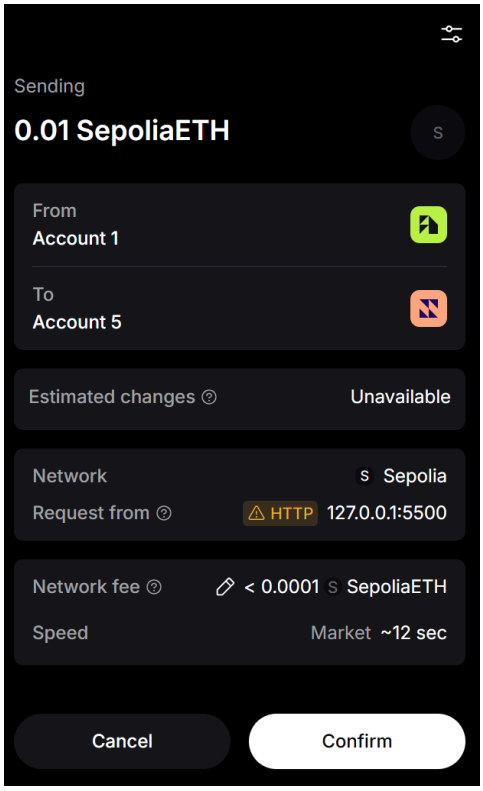
MetaMask connection request



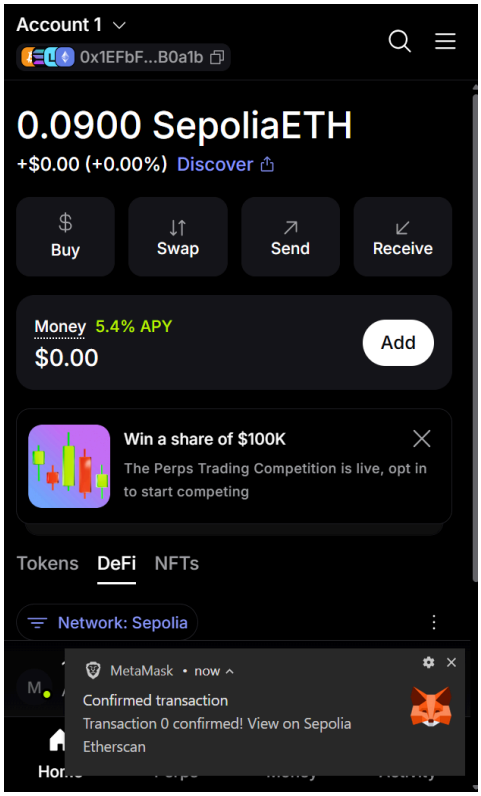
Successful wallet connection



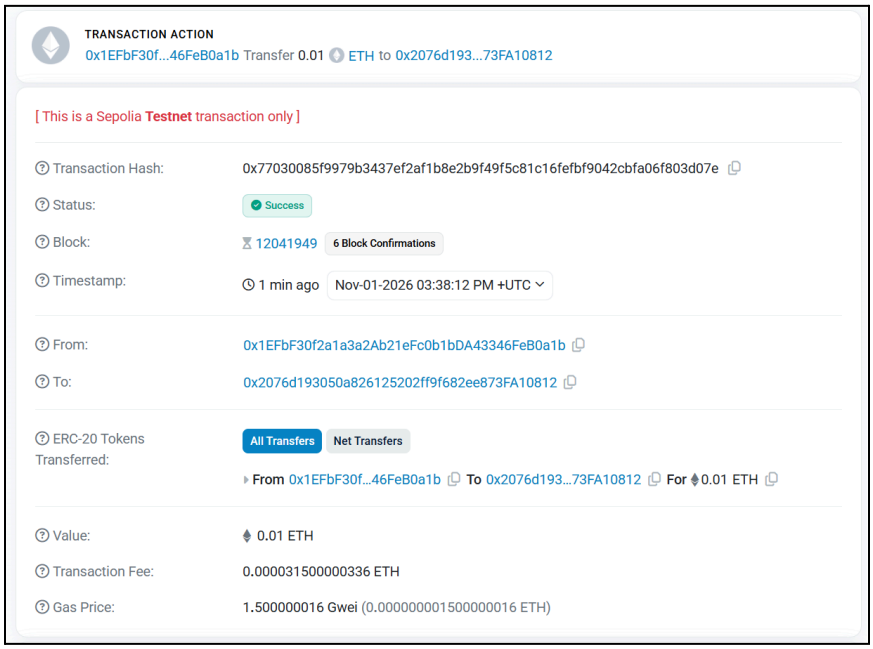
Transferring ETH to another account



Confirmation request for transfer



Confirmed Transaction and Sepholia sent



Transaction Details for Sepholia Etherscan

Conclusion

Thus, MetaMask was successfully configured and connected to the Ethereum Sepolia test network. A simple DApp was created and integrated with MetaMask using Ethers.js. ETH was successfully transferred between two accounts and verified using Sepolia Etherscan. The experiment demonstrated wallet–DApp integration and blockchain transaction processing